

Mismatch Matters: On-Policy Distillation Beyond Token Agreement

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Abstract

On-policy distillation (OPD) has emerged as a core component of modern LLM post-training pipelines, yet we reveal a failure mode: *degenerate agreement*, where students exploit repetitive loops to achieve near-perfect token agreement with the teacher despite globally flawed responses¹. We therefore shift our focus from agreement to teacher–student mismatch and find that the mismatch tokens can be mainly categorized to two types: student-excess tokens and student-deficit tokens. Specifically, student-excess tokens are those generated by the student but assigned near-zero probability by the teacher; their log-ratio corrections grow unbounded and destabilize the update. Student-deficit tokens, in contrast, are preferred by the teacher but rarely sampled by the student; their absence blocks the transfer of the teacher’s reasoning patterns. To tackle these mismatch directions, we propose **TIDE** (Token-level Independent Deficit–Excess correction), which applies bounded Hellinger shaping to suppress the most severe sampled excesses and an analytic teacher top- K injection to restore deficient probability mass without requiring deficit tokens to be sampled. Across mathematical reasoning benchmarks with multiple Qwen3 teacher–student pairs, TIDE consistently outperforms standard OPD and recent token-selection and reward-shaping baselines. Besides, the gains of TIDE are more pronounced under strong teacher–student mismatch, where it improves Avg@8 from 6.9% to 20.3%, reduces average response length by a factor of 3.6, and substantially reduces formatting failures. Our code is available at <https://github.com/yzc-666/TIDE>.

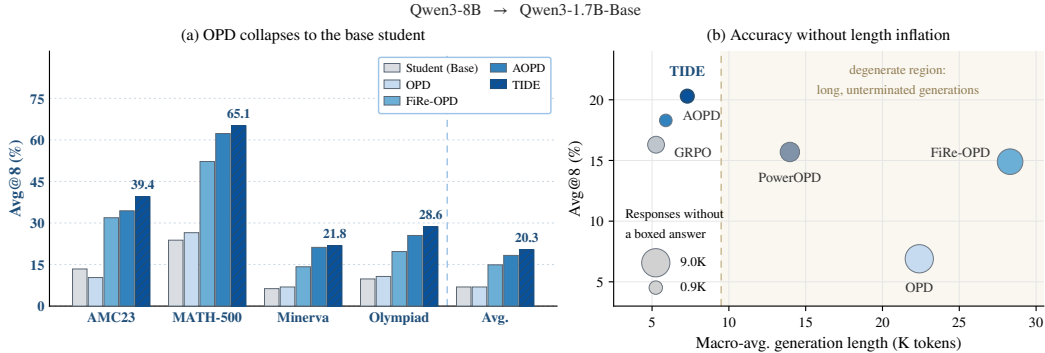


Figure 1: **TIDE achieves the highest accuracy without length inflation.** (a) TIDE consistently outperforms the baselines in Avg@8. (b) Unlike OPD and FiRe-OPD, TIDE maintains short, well-formed generations.

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¹Representative examples of degenerate agreement are provided in Section E.

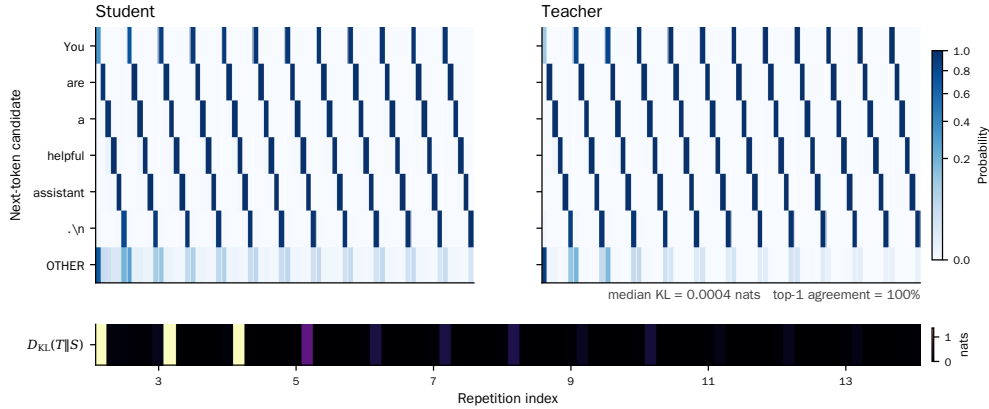


Figure 2: **Repetition creates degenerate agreement.** Visualization of a representative OPD rollout in which the student repeatedly generates the sequence `You are a helpful assistant. \n`. The left and right heatmaps show the student and teacher next-token distributions conditioned on progressively longer repetitive prefixes; rows correspond to tokens in the repeating unit, and darker colors indicate higher probability. As repetition continues, both models concentrate on the same periodic continuation: their top-1 predictions agree at 100%, while the median teacher–student KL falls to 0.0004 nats, as shown in the bottom strip. The globally degenerate rollout therefore appears locally well-aligned under token-level supervision.

1 Introduction

On-policy distillation (OPD) (Lu & Lab, 2025) has emerged as a powerful and widely adopted paradigm for post-training large language models (LLMs) (Agarwal et al., 2024; Gu et al., 2024), and has become a standard component in recent open-weight frontier models such as Qwen3, MiMo-V2-Flash, GLM-5, and Kimi K3 (Yang et al., 2025; Xiao et al., 2026; Zeng et al., 2026; Team et al., 2026). Rather than relying on fixed expert demonstrations, OPD samples trajectories from the student policy and queries an external teacher at the states the student actually visits. This on-policy supervision mitigates the exposure bias of supervised fine-tuning (SFT) (Bengio et al., 2015; Ross et al., 2011; Agarwal et al., 2024), while offering denser token-level feedback than the sparse outcome rewards used in reinforcement learning with verifiable rewards (RLVR) (Shao et al., 2024; Guo et al., 2025; Yu et al., 2026).

Despite these advantages, OPD is prone to abrupt length inflation, with student generations drifting toward the decoding cap (Luo et al., 2026). Our diagnostics trace this to persistent repetitive loops (see detailed analysis in Section 2.2 and one example in Fig. 2): the student repeatedly generates an identical prefix `"You are a helpful assistant. \n"`; conditioned on the student-generated prefix, the teacher finds each additional repetition increasingly predictable, until the two distributions (generated by student and teacher) nearly coincide (median KL 0.0004 nats). Critically, the student can thus drive teacher–student divergence to near zero while the overall response remains globally degenerate, revealing that high token-level agreement can actively mask global degeneration. We term this phenomenon *degenerate agreement*, and the underlying mechanism *student-induced teacher hacking*: repetition renders a degenerate continuation locally consistent to the teacher. This also resembles the low-KL trap recently observed for corrupted student prefixes (Xin et al., 2026).

A prevalent principle in recent OPD research is that training should focus on high teacher–student overlap to obtain valuable supervision. This assumption underpins recent work on token selection (Li et al., 2026a; Zhao et al., 2026b; Yang et al., 2026b), which treats high-overlap tokens as the most useful learning signals. However, this overlap-based criterion may also suffer from the risk of degenerate agreement: repetitive generations can produce artificially high overlap and then the teacher signal offers little new information. In such cases, high agreement ceases to be a reliable indicator of supervisory quality. This motivates us to shift the focus from agreement to mismatch (or

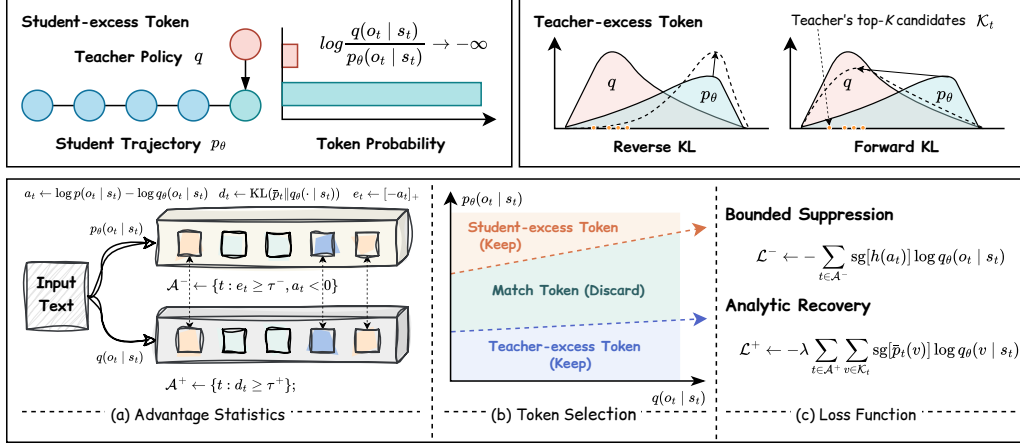


Figure 3: **An overview of TIDE pipeline.** Student-excess tokens are readily observed in student rollouts, but can produce unbounded negative advantages when assigned near-zero probability by the teacher. In contrast, student-deficit tokens are unlikely to be sampled and are therefore identified analytically using forward KL over the teacher’s top- K candidates. TIDE computes branch-specific mismatch statistics (a), independently retains positions with severe excess or deficit while discarding low-mismatch positions (b), and applies bounded suppression to excess tokens and analytic probability-mass recovery to deficit tokens (c).

disagreement) as the basis for supervision, and explore divergent tokens as an alternative, potentially more informative source of learning signal.

Concretely, we find that two types of mismatch tokens are rather distinct and important (see Section 2.1 for details): student-excess and student-deficit tokens. In particular, *Student-excess tokens* are overproduced by the student relative to the teacher; they are readily observed in rollouts, yet their log-ratio difference, can grow unbounded as the teacher probability approaches zero, destabilizing updates. For instance, in our OPD training, the most negative 1% of tokens contribute nearly half of the total gradient. *Student-deficit tokens*, conversely, are favored by the teacher but underweighted by the student. These tokens have a median probability of just 2.04×10^{-4} ; under our four-rollout budget, 95.5% of them are sampled with less than 1% probability. Despite this inaccessibility, we find that these deficit tokens carry rather valuable supervisory signal missed by standard on-policy supervision (we perform a simple deficit-only intervention and raise Avg@8 from 6.87 to 18.32 (see Section C.3)).

Based on our findings on the token-level mismatch, we propose **TIDE** (Token-level Independent Deficit–Excess correction), a correction mechanism that deliberately leverages mismatch tokens to achieve more informative and stable on-policy distillation (OPD) training. For student excess tokens, TIDE replaces the unbounded log-ratio with a principled Hellinger-shaped weight derived from the squared Hellinger divergence, which enforces bounded suppression without heuristic clipping or allowing extreme tokens to dominate the update. For student deficit tokens, TIDE introduces an analytic teacher top- K objective to restore missing probability mass directly, bypassing the need for rare rollouts. At each mini-batch, TIDE independently evaluates both mismatch directions and selectively applies the corresponding correction only to the most severe positions, discarding all neutral tokens from the loss. We further develop theoretical results demonstrating that the excess branch provides bounded, locally faithful corrections aligned with a proper divergence geometry, while the deficit branch directly addresses an irreducible coverage gap that sampling-based methods cannot resolve. Empirical evaluations on mathematical reasoning benchmarks and various teacher–student pairs (including Qwen3 family models (Yang et al., 2025)) confirm that TIDE consistently outperforms strong baselines, with ablation studies further validating the complementary roles of its two correction components. Our contributions are summarized as follows.

- We identify *student-induced teacher hacking* as a previously overlooked failure mode of OPD. Once the student enters a repetitive prefix, the teacher can favor the same continuation, creating a low-KL matched region in which degeneration receives little corrective signal.

- We reveal an intrinsic asymmetry in informative teacher–student disagreement. Student-excess tokens are readily observed but can induce unstable updates, whereas student-deficit tokens carry valuable teacher information but are rarely observed in student rollouts.
- We introduce TIDE, which combines quantile-gated Hellinger suppression for student excess with analytic teacher-top- K recovery for student deficit while removing redundant supervision from matched tokens. TIDE consistently outperforms strong OPD baselines; notably, for our Qwen experiment setting, it raises Avg@8 from 6.87 to 20.20 while reducing the average response length from 22,395 to 7,294 tokens.

2 Problem Setup and Empirical Observations

2.1 Problem Setup

We begin by establishing the basic setup. Let $\mathcal{D}$ denote the distribution over input prompts x , and let $\mathcal{V}$ be the vocabulary. At a given state s , the fixed teacher and the trainable student define next-token distributions $p(\cdot|s)$ and $q_\theta(\cdot|s)$, respectively, where θ denotes the student parameters. Since both models use a softmax with full support over the vocabulary, these probabilities are strictly positive for every token $v \in \mathcal{V}$. For a prompt $x \sim \mathcal{D}$, the student autoregressively generates a response $o = (o_1, \dots, o_T) \sim q_\theta(\cdot|x)$, where T denotes the total response length, including the end-of-sequence token. At each position t , the state visited by the student is $s_t = (x, o_{<t})$. The corresponding trajectory likelihoods factorize autoregressively in the usual way: for instance, $q_\theta(o|x) = \prod_{t=1}^T q_\theta(o_t|s_t)$, and analogously for the teacher $p(o|x)$.

The objective of OPD is to minimize the expected trajectory-level reverse KL divergence:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(q_\theta(\cdot|x) \| p(\cdot|x))] = \mathbb{E}_{\substack{x \sim \mathcal{D} \\ o \sim q_\theta(\cdot|x)}} \left[\sum_{t=1}^T \log \frac{q_\theta(o_t|s_t)}{p(o_t|s_t)} \right], \quad (1)$$

where D_{KL} is the Kullback–Leibler divergence.

Following recent practice (Lu & Lab, 2025), we implement this objective using a detached token-level policy-gradient estimator (Schulman et al., 2017):

$$g_{\text{OPD}}(\theta) = -\mathbb{E} \left[\sum_{t=1}^T \text{sg}[a_t] \nabla_\theta \log q_\theta(o_t|s_t) \right], \quad a_t = \log \frac{p(o_t|s_t)}{q_\theta(o_t|s_t)}.$$

Here the expectation is taken over the same prompts and student rollouts as in Equation 1, and sg denotes the stop-gradient operation. We interpret the scalar a_t as the advantage of the sampled token at step t : a positive value increases its likelihood, while a negative one decreases it. A crucial subtlety is that this update rule only directly affects tokens actually drawn from the student’s current distribution, i.e., a dependence that becomes especially consequential when the teacher and student predictions diverge (see more discussion in Section 2.3).

2.2 Motivation: Student-Induced Teacher Hacking

In this section, we highlight several key observations about the interaction between student-generated trajectories and teacher supervision during OPD; the experimental setup is detailed in Section 4.1. In OPD, the teacher evaluates each token conditioned on the prefix generated by the student, rather than their own original rollout alone. This allows the student to influence the teacher’s conditional predictions by steering the context, which can lead to an exploitable failure mode: the student can reduce the measured teacher–student divergence by generating contexts that make the teacher’s outputs more aligned with its own, without actually improving its reasoning ability. We refer to this phenomenon as *student-induced teacher hacking*.

We first examine the trajectory of the OPD training and find a conspicuous pattern *repetitive generation*: as training proceeds, the student increasingly falls into loops where the same token sequence is repeated many times, e.g., repeating `I hope it is correct` 149 times in math problems. We then evaluate a fixed set of prompts across training checkpoints and find that the proportion of repetitive rollouts rises from 16.8% to 48% (Fig. 4a). These loops are rarely transient; once the

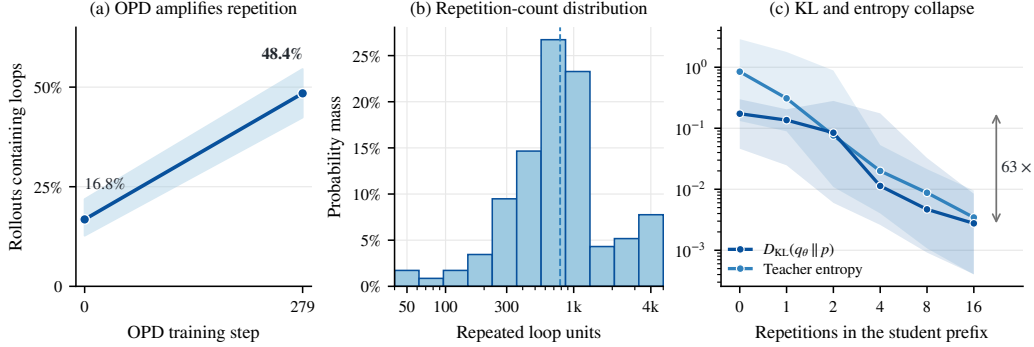


Figure 4: **Repetition emerges as an exploit of teacher conditioning in OPD.** (a) On a fixed set of DAPO-Math-17K prompts, the fraction of rollouts containing repetitive loops increases from 16.8% before training to 48.4% after OPD training; shaded regions denote 95% confidence intervals. (b) Repetitive rollouts from the final checkpoint exhibit a heavy-tailed distribution of repetition counts. (c) Increasing the number of repetitions in a controlled student prefix simultaneously reduces student–teacher KL and teacher entropy, with sixteen repetitions lowering KL by $63\times$.

student enters one, it tends to persist for many iterations, yielding the heavy-tailed distribution of repetition counts shown in Fig. 4b. Notably, neither the prompts nor the objective function explicitly encourage repetition. This behavior is conjectured to emerge organically from the interplay between OPD optimization and the student-conditioned teacher signal.

We next examine how the teacher reacts once the student falls into such a loop. For each naturally occurring repetitive rollout, we isolate its repeating unit and construct controlled prefixes with varying numbers of repetitions, then evaluate all prefixes on the same set of subsequent tokens. As shown in Fig. 4c, increasing the repetition count simultaneously reduces both the student–teacher KL divergence and the teacher’s predictive entropy. With sixteen repetitions, the KL drops by a factor of $63\times$. Far from correcting the repetition, the teacher grows increasingly confident in continuing it. This pattern is consistent under greedy decoding (see Section C.1): when prompted with natural loop prefixes from the student, the teacher continues the same repetitive pattern in 93% of cases. In effect, repetition turns a degenerate student trajectory into apparent teacher–student agreement, precisely eroding the supervisory signal where it is most needed.

2.3 Informative mismatch is directional.

The results above show that teacher–student agreement may not necessarily constitute useful supervision: the two models can agree even on degenerate trajectories induced by the student. Our matched-only control experiment provides further evidence for this distinction: supervising only the 60% best-matched positions raises the Avg@8 of Qwen3-1.7B-Base from 6.87 to just 7.18, whereas supervising the 20% most mismatched positions more than doubles it to 14.58 (see Section C.2 and Table 6). Training on matched positions yields substantially smaller gains than supervising a much smaller set of mismatched positions, particularly under strong teacher–student mismatch. These observations motivate us to shift the gradient budget away from already matched positions and toward states where the two distributions meaningfully disagree.

Having established where supervision should be concentrated, we next consider how different forms of disagreement should be handled. The answer follows from the asymmetric structure of the reverse-KL objective used by vanilla OPD in Equation 1. This divergence is well known to be exclusive and zero-forcing: it aggressively penalizes the student for assigning probability to tokens that the teacher deems unlikely, yet offers comparatively weak pressure to cover teacher-preferred tokens that fall outside the student’s effective support (Minka et al., 2005; Gu et al., 2024; Agarwal et al., 2024). These two sides of reverse KL introduce a directional asymmetry to teacher–student mismatch in OPD. On one hand, when the student over-weights a token that the teacher assigns low probability, the discrepancy is easy to observe through sampling, but suppressing it can lead to unstable updates. On the other hand, when the student under-weights a token that the teacher favors,

the resulting supervision may be highly informative, yet it remains largely inaccessible to standard student sampling. The two directions are visualized in the top row of Fig. 3: student excess (top left) is readily observable but can induce unstable sampled-token corrections, whereas student deficit (top right) is difficult to observe and requires analytic teacher-top- K guidance.

This asymmetry is especially pronounced in sampled-token OPD. A negative advantage $a_t < 0$ indicates that the student overweights the sampled token relative to the teacher, and on-policy rollouts expose such tokens readily, since sampling follows the student’s own distribution. The suppression magnitude $-a_t = \log[q_\theta(o_t|s_t)/p(o_t|s_t)]$ measures this gap: it stays near zero when both models consider a token similarly unlikely, but grows without bound as the teacher probability vanishes while the student retains mass (Zhao et al., 2026a). A handful of strongly teacher-rejected tokens can thus dominate the reverse-KL update and destabilize training. Our first branch locates the largest gaps along the sampled trajectory and suppresses them with a bounded signal in place of the raw log-ratio. This observable-but-unstable regime corresponds to the student-excess case in Fig. 3 (top left).

The opposite direction, however, cannot be addressed by simply rescaling or reshape a_t . Consider a token v that the teacher assigns high probability but the student deems nearly impossible. Its sampling-weighted teacher-favoring signal satisfies

$$q_\theta(v|s) \log \frac{p(v|s)}{q_\theta(v|s)} \longrightarrow 0 \quad \text{as } q_\theta(v|s) \rightarrow 0.$$

Thus, the more severely the student under-weights a teacher-preferred token, the less likely OPD is to ever sample it, and hence the less opportunity the algorithm has to reinforce it. Our empirical measurements confirm that these missing modes are not only difficult to observe through standard sampling, but also carry substantial learning value when recovered (Section C.3). This motivates our second branch: rather than waiting for teacher-preferred tokens to be sampled by chance, we directly inspect the teacher’s top- K support and provide explicit analytic guidance toward these under-covered tokens. This hard-to-sample regime corresponds to the student-deficit case in Fig. 3 (top right), where the teacher’s top- K candidates expose alternatives that may never appear in the student rollout.

The directional asymmetry of reverse KL thus calls for a solution that treats the two directions differently. Student-overweighted tokens are easy to spot but require stable suppression; teacher-preferred tokens are rarely seen and need explicit support recovery. The next section turns these two observations into separate suppression and recovery branches.

3 Methodology

In this section, we present TIDE, an on-policy distillation method that uses the magnitude of teacher–student mismatch to determine *where* to update and its direction to determine *how* to update. At each student-visited state, TIDE diagnoses teacher–student mismatch from two complementary perspectives: one flags tokens where the student over-estimates probability, and the other detects teacher-preferred tokens that the student rarely generates. Only positions exhibiting substantial mismatch in either direction receive corrective updates; all remaining tokens are excluded from the distillation objective. The overall procedure is illustrated in Fig. 3. Its bottom row summarizes three stages: TIDE computes branch-specific mismatch statistics (bottom left), independently routes severe positions to the two branches (bottom center), and applies the corresponding correction objectives (bottom right).

3.1 Stabilizing Student-Excess Updates

We first consider positions where the sampled token receives more probability from the student than from the teacher, i.e., student-excess tokens. These positions satisfy $a_t < 0$ and can be ranked by the magnitude of the excess $e_t = [-a_t]_+$, where $[u]_+ = \max\{u, 0\}$.

Let $\mathcal{B}$ denote the set of response positions in a training batch. To better locate the tokens with severe mismatch, we consider a constant fraction ρ^- of positions ranked by their excess scores. Specifically, we define the corresponding batch-quantile threshold and selection mask as

$$\tau^- = Q_{1-\rho^-}(\{e_j : j \in \mathcal{B}\}), \quad m_t^- = \mathbb{1}[a_t < 0, e_t \geq \tau^-],$$

where Q_α denotes the empirical α -quantile. This selection focuses the distillation signal on positions where the teacher assigns low probability to the student’s sampled token. The score-and-gate construction is shown by the orange route from the bottom-left statistics panel to the retained student-excess region in the bottom-center panel of Fig. 3.

As discussed above, the teacher assigns substantially less probability than the student to tokens selected by the excess gate, making a_t strongly negative. The resulting log-ratio penalty $-a_t = \log[q_\theta(o_t|s_t)/p(o_t|s_t)]$ grows without bound as the teacher probability approaches zero. Thus, directly applying the original OPD advantage after selection would concentrate the gradient too heavily on a small number of extreme corrections, leading to the training instability.

We address this problem by applying a bounded, monotonic transformation to the raw advantage at each selected position. Clipping-style functions (e.g., hard clipping or tanh) would also bound the update, but they rely on an arbitrary saturation scale, collapse all severe discrepancies to nearly identical corrections, and yield an update that no longer corresponds to any well-defined divergence between the two distributions. We instead adopt the Hellinger-shaped function, which bounds the update and locally preserves the log-ratio while remaining the exact gradient of a proper divergence:

$$h(a_t) = 2[\exp(a_t/2) - 1].$$

Accordingly, we take the following the policy gradient in OPD:

$$g_{\text{exc}} = -\mathbb{E} \left[\sum_t m_t^- \text{sg}[h(a_t)] \log q_\theta(o_t|s_t) \right].$$

Thus, quantile selection determines where suppression is informative, while Hellinger shaping prevents the selected reverse-KL tail from destabilizing optimization. The resulting bounded suppression objective appears in Fig. 3 (bottom right).

The following proposition explains the critical properties of Hellinger transformation.

Proposition 1 (Bounded, locally faithful Hellinger shaping). *For a student-excess token, $a < 0$,*

$$-2 < h(a) < 0, \quad h(a) = a + O(a^2) \quad \text{as } a \rightarrow 0.$$

Moreover, before position selection, a token sampled from the student gives the following unbiased gradient identity when $h(a(v))$ is treated as a detached coefficient:

$$\mathbb{E}_{v \sim q_\theta} [-h(a(v)) \nabla_\theta \log q_\theta(v|s)] = 4 \nabla_\theta H^2(p, q). \quad (2)$$

Proposition 1 explains the advantage of Hellinger shaping compared to the simple clipping of the raw advantage. Its negative side is bounded, preventing teacher-rejected tokens from producing arbitrarily large suppression updates. At the same time, $h(a) \approx a$, i.e., in the regime near agreement, so moderate corrections retain the first-order behavior of standard OPD. Moreover, Equation 2 shows that this transformation arises naturally from a proper divergence, grounding the design in a principled objective rather than an ad hoc heuristic.

3.2 Recovering Under-Covered Teacher Support

Unlike the excess case, missing teacher support cannot be detected from the sampled advantage a_t , which only compares the two distributions at the student’s chosen token. To identify teacher-preferred tokens that the student overlooks, we inspect the teacher’s top- K support. Define the top- K support and the renormalized teacher distribution within this support as follows:

$$\mathcal{K}_t = \text{TopK}_{v \in \mathcal{V}} p(v|s_t); \quad \bar{p}_t(v) = \frac{p(v|s_t)}{\sum_{u \in \mathcal{K}_t} p(u|s_t)}, \quad v \in \mathcal{K}_t.$$

Then we quantify the student’s deficit on this support based on the following metric

$$d_t = \sum_{v \in \mathcal{K}_t} \bar{p}_t(v) \log \frac{\bar{p}_t(v)}{q_\theta(v|s_t)}. \quad (3)$$

This is the blue deficit statistic shown in the bottom-left panel of Fig. 3, motivated by the teacher-preferred alternatives in the top-right panel. Notably, unlike standard top- K OPD (Jin et al., 2026),

we do not renormalize the student probabilities over $\mathcal{K}_t$. The score d_t quantifies how well the student covers the teacher’s preferred continuations, and enjoys a natural interpretation: it is nonnegative, vanishes only when the student matches $\bar{p}_t$ within $\mathcal{K}_t$ and assigns all its probability mass to this set, and grows large whenever the student underweights tokens that the teacher deems plausible. These properties are formalized in [Proposition 2](#) below.

Proposition 2 (Deficit-score decomposition). *Let $Q_t = \sum_{v \in \mathcal{K}_t} q_\theta(v|s_t)$ denote the total student mass on $\mathcal{K}_t$, and let $q_t^K(v) = q_\theta(v|s_t)/Q_t$ be the corresponding renormalized distribution for $v \in \mathcal{K}_t$. Then the deficit score in Equation 3 satisfies*

$$d_t = \underbrace{D_{\text{KL}}(\bar{p}_t \| q_t^K)}_{\text{mismatch within } \mathcal{K}_t} + \underbrace{(-\log Q_t)}_{\text{insufficient mass on } \mathcal{K}_t} \geq 0.$$

Equality holds if and only if $q_t^K = \bar{p}_t$ and $Q_t = 1$.

Additionally, this decomposition reveals two complementary failure modes: the KL term captures misallocation of probability among the teacher’s leading candidates, while $-\log Q_t$ measures how much student mass falls outside this set. This also justifies why we do not renormalize the student distribution in Equation 3: renormalization would eliminate the coverage gap encoded by $-\log Q_t$, making the score blind to the very deficiency we aim to correct.

A large d_t signals that the student poorly covers the teacher’s preferred continuations at that state. We therefore use d_t as a priority score and concentrate the deficit branch on the most severe positions. Within each batch, we retain the largest ρ^+ fraction of scores:

$$\tau^+ = \text{Quantile}_{1-\rho^+}(\{d_j : j \in \mathcal{B}\}), \quad m_t^+ = \mathbb{1}[d_t \geq \tau^+],$$

and denote the active set as $\mathcal{A}^+ = \{t : m_t^+ = 1\}$. For these selected positions, we do not rely on sampling to surface the missing tokens. Instead, we directly distill the teacher’s renormalized top- K distribution:

$$\mathcal{L}_{\text{def}} = -\mathbb{E} \left[\sum_t m_t^+ \sum_{v \in \mathcal{K}_t} \bar{p}_t(v) \log q_\theta(v|s_t) \right]. \quad (4)$$

This completes the blue route in [Fig. 3](#): the deficit score selects positions in the bottom-center panel, and the analytic top- K objective performs probability-mass recovery in the bottom-right panel. The underlying principle is that student rollouts determine where to learn, but not which tokens to learn: the deficit branch uses the student’s mass on the teacher’s top- K set to detect missing support and directly distills those candidates when coverage is insufficient.

3.3 Joint Objective

The excess and deficit branches operate on the same student-generated states but address complementary forms of mismatch. The two routing decisions in [Fig. 3](#) (bottom center) are independent: a position may activate the excess branch, the deficit branch, or both, while positions selected by neither branch receive no distillation update. We optimize the two branches jointly as

$$\mathcal{L}_{\text{TIDE}} = \mathcal{L}_{\text{exc}} + \lambda \mathcal{L}_{\text{def}}, \quad (5)$$

where λ controls the strength of teacher-support recovery relative to excess suppression. Each branch contributes only at the positions selected by its corresponding gate; positions selected by neither branch receive no distillation update. [Algorithm 1](#) summarizes the resulting training procedure.

4 Experiments

4.1 Experimental Setup

Models and baselines. We employ two teacher–student pairs with different degrees of model mismatch. The first uses JustRL-DeepSeek-1.5B ([He et al., 2025](#)) as the teacher and DeepSeek-R1-Distill-Qwen-1.5B ([Guo et al., 2025](#)) as the student, while the second uses Qwen3-8B as the teacher

Algorithm 1: TIDE Training

Input: prompt distribution $\mathcal{D}$, student q_θ , teacher p , top- K size K , keep ratios ρ^- , ρ^+ , recovery weight λ , learning rate η

Output: Trained student parameters θ

```

1 while not converged do
2   Sample a prompt batch  $x \sim \mathcal{D}$  and generate  $o \sim q_\theta(\cdot|x)$ 
3   Let  $\mathcal{B}$  be all valid response-token positions and  $N = |\mathcal{B}|$ 
4   foreach  $t \in \mathcal{B}$  do
5     Construct  $s_t = (x, o_{<t})$  and compute  $a_t \leftarrow \log p(o_t|s_t) - \log q_\theta(o_t|s_t)$ ,  $e_t \leftarrow [-a_t]_+$ 
6     Obtain  $\mathcal{K}_t \leftarrow \text{TopK}_v p(v|s_t)$  and normalize  $\bar{p}_t(v) \leftarrow p(v|s_t) / \sum_{u \in \mathcal{K}_t} p(u|s_t)$ 
7     Compute  $d_t \leftarrow \sum_{v \in \mathcal{K}_t} \bar{p}_t(v) \log \frac{\bar{p}_t(v)}{q_\theta(v|s_t)}$ 
8   Set  $\tau^- \leftarrow Q_{1-\rho^-}(\{e_t : t \in \mathcal{B}\})$  and  $\tau^+ \leftarrow Q_{1-\rho^+}(\{d_t : t \in \mathcal{B}\})$ 
9   Select  $\mathcal{A}^- \leftarrow \{t : a_t < 0, e_t \geq \tau^-\}$  and  $\mathcal{A}^+ \leftarrow \{t : d_t \geq \tau^+\}$ 
10  Compute
      
$$\mathcal{L}_{\text{exc}} \leftarrow -\frac{1}{N} \sum_{t \in \mathcal{A}^-} \text{sg}[2(e^{a_t/2} - 1)] \log q_\theta(o_t|s_t)$$

11  Compute
      
$$\mathcal{L}_{\text{def}} \leftarrow -\frac{1}{N} \sum_{t \in \mathcal{A}^+} \sum_{v \in \mathcal{K}_t} \bar{p}_t(v) \log q_\theta(v|s_t)$$

12  Set  $\mathcal{L}_{\text{TIDE}} \leftarrow \mathcal{L}_{\text{exc}} + \lambda \mathcal{L}_{\text{def}}$ 
13  Update  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{TIDE}}$ 
14 return  $\theta$ 

```

and Qwen3-1.7B-Base as the student (Yang et al., 2025). Following Li et al. (2026a), we characterize the initial teacher–student mismatch using the overlap between their top- K candidate sets on student-generated prefixes. A higher initial overlap indicates a smaller teacher–student mismatch, while a lower overlap indicates a larger mismatch. Under this criterion, the JustRL-DeepSeek-1.5B $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B pair constitutes our *weak-mismatch* setting, while the Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base pair constitutes our *strong-mismatch* setting.

We compare TIDE against GRPO (Shao et al., 2024), standard OPD (Lu & Lab, 2025), FiRe-OPD (Li et al., 2026b), PowerOPD (Zhao et al., 2026a), and AOPD (Jia et al., 2026). PowerOPD applies a bounded power transformation to stabilize the sampled-token reward, whereas AOPD preserves positive reinforcement while replacing non-positive updates with localized divergence minimization. We additionally report the original teacher and student models as reference points.

Training Details. We train all methods for one epoch on DAPO-Math-17K (Yu et al., 2026) using the verl framework (Sheng et al., 2024). Each prompt produces four on-policy student responses, and all methods use the same student initialization, data order, rollout budget, and optimizer configuration. For TIDE, we set $K = 16$, $\rho^- = \rho^+ = 0.2$, and $\lambda = 1.0$. Full training, implementation, baseline, and evaluation details are provided in Section D.

4.2 Main Results

Overall Performance. Tables 1 and 2 reveal a clear mismatch-dependent pattern. Under weak teacher–student mismatch, the distillation methods achieve broadly comparable Avg@8, while TIDE obtains the highest overall Avg@8 and Pass@8 of 46.7% and 65.0%, improving over OPD by 1.0 and 3.2 points, respectively. The differences become substantially larger under strong mismatch. Standard OPD falls to the original student’s Avg@8 level of 6.9%, whereas TIDE reaches 20.3%, outperforming OPD, FiRe-OPD, GRPO, PowerOPD, and AOPD by 13.4, 5.4, 4.0, 4.6, and 2.0 points, respectively. For Pass@8, TIDE obtains 34.2%, compared with 25.3% for OPD, 27.2% for FiRe-OPD, 29.5% for GRPO, 31.2% for PowerOPD, and 34.0% for AOPD. Moreover, TIDE achieves the best-or-tied Avg@8 on seven of nine benchmarks and the best-or-tied Pass@8 on six of nine benchmarks under strong mismatch. These results are consistent with our analysis: sampled-token corrections are often sufficient when the teacher and student distributions substan-

Table 1: **Avg@8 results across different teacher–student mismatch settings (%)**. Each model generates eight responses per problem. *Student* and *Teacher* denote reference models and are excluded from bolding. **Bold** denotes the best trained model in each column. Δ denotes the improvement of TIDE over OPD.

Method	AIME'24	AIME'25	AMC23	MATH	Minerva	Olymp.	BRUMO	CMIMC	HMMT	Avg.
<i>Weak mismatch: JustRL-DeepSeek-1.5B $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B</i>										
Student (Base)	27.1	24.2	73.1	83.4	28.0	44.3	27.1	14.4	12.1	37.1
Teacher	53.3	40.0	87.5	86.3	33.5	54.0	43.3	22.8	22.1	49.2
GRPO	22.5	21.2	72.5	83.9	29.5	46.3	33.3	16.6	14.2	37.8
OPD	43.8	32.1	82.2	86.2	34.1	51.6	40.4	21.9	19.2	45.7
FiRe-OPD (Li et al., 2026b)	39.2	37.1	84.7	85.3	33.2	52.3	42.5	21.6	23.3	46.6
PowerOPD (Zhao et al., 2026a)	60.0	31.2	81.6	84.0	32.6	50.6	38.3	19.7	20.4	46.5
AOPD (Jia et al., 2026)	50.8	33.3	81.9	85.4	33.0	51.3	41.2	20.6	17.9	46.2
TIDE	45.8	33.3	84.4	86.7	33.5	52.5	42.9	22.2	19.2	46.7
Δ vs. OPD	+2.0	+1.2	+2.2	+0.5	−0.6	+0.9	+2.5	+0.3	0.0	+1.0
<i>Strong mismatch: Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base</i>										
Student (Base)	3.3	0.0	13.4	23.8	6.3	9.8	5.0	0.0	0.0	6.9
Teacher	23.3	23.3	68.8	83.3	28.0	49.8	29.2	11.6	11.7	36.6
GRPO	3.3	0.4	35.0	55.5	18.7	23.3	9.6	0.9	0.0	16.3
OPD	2.9	1.2	10.3	26.5	6.9	10.7	2.9	0.3	0.0	6.9
FiRe-OPD (Li et al., 2026b)	3.3	4.2	31.9	52.2	14.2	19.7	7.1	0.9	0.8	14.9
PowerOPD (Zhao et al., 2026a)	6.7	4.2	26.2	55.1	15.1	24.5	7.1	1.9	0.4	15.7
AOPD (Jia et al., 2026)	3.3	4.6	34.4	62.3	21.2	25.5	10.8	1.9	0.4	18.3
TIDE	10.0	4.6	39.4	65.1	21.8	28.6	10.4	2.8	0.4	20.3
Δ vs. OPD	+7.1	+3.4	+29.1	+38.6	+14.9	+17.9	+7.5	+2.5	+0.4	+13.4

Table 2: **Pass@8 results across different teacher–student mismatch settings (%)**. Pass@8 is the percentage of problems for which at least one of eight generated responses is correct. *Student* and *Teacher* denote reference models and are excluded from bolding. **Bold** denotes the best trained model in each column. Δ denotes the improvement of TIDE over OPD.

Method	AIME'24	AIME'25	AMC23	MATH	Minerva	Olymp.	BRUMO	CMIMC	HMMT	Avg.
<i>Weak mismatch: JustRL-DeepSeek-1.5B $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B</i>										
Student (Base)	36.7	33.3	95.0	95.0	43.4	60.8	56.7	30.0	20.0	52.3
Teacher	53.3	40.0	92.5	94.2	45.2	68.2	66.7	42.5	36.7	59.9
GRPO	43.3	36.7	90.0	94.4	45.6	64.8	63.3	32.5	30.0	55.6
OPD	76.7	53.3	95.0	94.6	44.9	66.8	56.7	35.0	33.3	61.8
FiRe-OPD (Li et al., 2026b)	80.0	56.7	95.0	94.2	44.1	68.0	60.0	35.0	43.3	64.0
PowerOPD (Zhao et al., 2026a)	60.0	43.3	97.5	94.4	46.3	67.2	60.0	32.5	33.3	59.4
AOPD (Jia et al., 2026)	73.3	43.3	95.0	94.8	46.0	67.4	63.3	40.0	26.7	61.1
TIDE	80.0	50.0	95.0	94.6	45.2	68.1	66.7	42.5	43.3	65.0
Δ vs. OPD	+3.3	−3.3	0.0	0.0	+0.3	+1.3	+10.0	+7.5	+10.0	+3.2
<i>Strong mismatch: Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base</i>										
Student (Base)	3.3	0.0	40.0	68.2	24.3	34.3	20.0	0.0	0.0	21.1
Teacher	33.3	43.3	92.5	94.2	39.7	67.7	60.0	32.5	26.7	54.4
GRPO	6.7	3.3	60.0	83.4	36.0	45.5	23.3	7.5	0.0	29.5
OPD	13.3	13.3	45.0	73.8	28.7	37.7	13.3	2.5	0.0	25.3
FiRe-OPD (Li et al., 2026b)	3.3	10.0	55.0	77.6	32.7	40.9	16.7	5.0	3.3	27.2
PowerOPD (Zhao et al., 2026a)	6.7	10.0	57.5	83.6	37.1	47.2	23.3	12.5	3.3	31.2
AOPD (Jia et al., 2026)	3.3	16.7	65.0	86.0	40.4	47.8	33.3	10.0	3.3	34.0
TIDE	16.7	10.0	60.0	86.2	41.9	50.3	26.7	12.5	3.3	34.2
Δ vs. OPD	+3.4	−3.3	+15.0	+12.4	+13.2	+12.6	+13.4	+10.0	+3.3	+8.9

tially overlap, but analytic recovery of teacher-preferred alternatives becomes increasingly important as mismatch grows and the student’s sampled support becomes less informative.

Generation Stability. Table 3 shows that the accuracy gap is accompanied by substantially different generation behavior. Under strong mismatch, OPD and FiRe-OPD produce 22,395- and 28,307-token responses, with 65.5% and 54.0% missing a boxed answer; PowerOPD reduces but does not eliminate this degeneration, averaging 13,972 tokens with a 26.3% missing-answer rate. In contrast,

Table 3: **Generation diagnostics across mismatch settings.** Avg@8 and response length are macro-averaged over nine benchmarks. Missing \boxed denotes the percentage of all generated responses that do not contain a boxed final answer. **Bold** denotes the best trained model in each row; response length is not bolded, since shorter generations are not inherently preferable.

Metric	GRPO	OPD	FiRe-OPD	PowerOPD	AOPD	TIDE
<i>Weak mismatch: JustRL-DeepSeek-1.5B $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B</i>						
Avg@8 (%)	37.8	45.7	46.6	46.5	46.2	46.7
Avg. length	8,048	7,657	8,055	8,658	7,982	7,571
Missing \boxed (%)	1.8	4.2	4.2	5.3	4.8	3.7
<i>Strong mismatch: Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base</i>						
Avg@8 (%)	16.3	6.9	14.9	15.7	18.3	20.3
Avg. length	5,261	22,395	28,307	13,972	5,894	7,294
Missing \boxed (%)	17.1	65.5	54.0	26.3	4.7	5.4

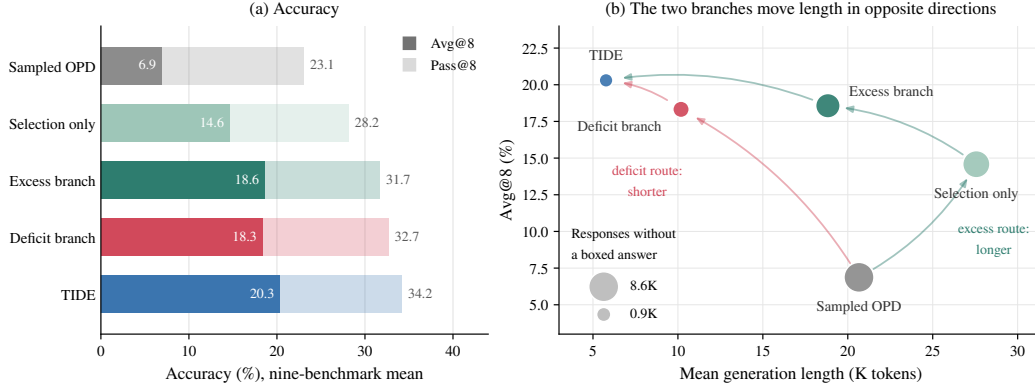


Figure 5: **Component ablation on Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base.** (a) Avg@8 (solid) and Pass@8 (shaded), macro-averaged over nine benchmarks. (b) Accuracy versus macro-averaged response length; marker area denotes the number of responses without a parseable final answer. The excess and deficit branches provide complementary improvements, and their combination achieves the highest accuracy with the shortest responses.

TIDE achieves the highest Avg@8 of 20.3% while reducing response length to 7,294 tokens and the missing-answer rate to 5.4%. This gain cannot be attributed to shorter responses alone: GRPO and AOPD generate even shorter outputs but remain 4.0 and 2.0 Avg@8 points behind TIDE, respectively.

4.3 Ablation Studies

Component Ablation. Table 4 and Fig. 5 shows that the three components play complementary roles. Selection alone raises Avg@8 from 6.87 to 14.58, confirming the value of concentrating supervision on informative positions. However, it also increases the macro-averaged response length from 22.4K to 29.7K tokens, indicating that selection alone does not prevent generation degeneration. Applying bounded Hellinger shaping to the selected excess positions further improves Avg@8 to 18.57 and partially controls the length increase, reducing it to 23.6K. Analytic deficit recovery independently reaches a similar Avg@8 of 18.32, while producing substantially shorter responses of 13.9K tokens. Combining all three components in TIDE achieves the best Avg@8 and Pass@8 of 20.34 and 34.17, respectively, while further reducing response length to 7.3K tokens. Overall, selection determines *where* to learn, Hellinger shaping stabilizes excess suppression, and deficit recovery restores teacher-preferred continuations that student sampling tends to miss.

Table 4: **Component ablation on Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base.** Avg@8, Pass@8, and response length are macro-averaged over nine benchmarks.

Method	Selection	Hellinger	Deficit	Avg@8	Pass@8	Length
Sampled OPD	–	–	–	6.87	25.29	22.4K
Selection only	✓	–	–	14.58	28.16	29.7K
Excess branch	✓	✓	–	18.57	31.72	23.6K
Deficit branch	–	–	✓	18.32	32.71	13.9K
TIDE	✓	✓	✓	20.34	34.17	7.3K

λ	Avg@8 $\uparrow$	Pass@8 $\uparrow$	Avg. Len. $\downarrow$	Format Err. $\downarrow$	Dist.-4 $\uparrow$
0.25	20.09	32.60	26.9K	5,510	0.145
0.5	20.35	36.68	20.0K	2,488	0.128
1.0	20.34	34.17	7.3K	706	0.305
2.0	19.20	35.41	5.1K	482	0.417

Table 5: Sensitivity to the deficit-recovery weight λ with fixed gates $\rho^- = \rho^+ = 0.2$. Avg@8, Pass@8, response length, and Distinct-4 are macro-averaged over nine benchmarks; format errors are totals.

Sensitivity to Hyperparameters. Table 5 studies the deficit-recovery weight λ with fixed gates $\rho^- = \rho^+ = 0.2$. Accuracy remains relatively stable around the default setting: $\lambda = 0.25, 0.5$, and 1.0 achieve similar Avg@8 scores. Their generation behavior, however, differs substantially. Reducing λ from 1.0 to 0.25 increases the macro-averaged response length from $7.3K$ to $26.9K$ tokens and format errors from 706 to $5,510$, indicating that insufficient deficit recovery allows degenerate continuations to persist. Increasing λ to 2.0 further shortens responses but reduces Avg@8 to 19.20 . We therefore use $\lambda = 1.0$ as a balanced default. Sensitivity to the gate keep-rate is reported in Table 8.

5 Conclusion

We identify *degenerate agreement* as a failure mode of on-policy distillation: local token-level matching can mask globally flawed responses. Teacher–student mismatch is directional: student-excess tokens induce unstable corrections, whereas valuable student-deficit tokens are rarely sampled. TIDE addresses both with bounded excess correction and analytic deficit recovery. Across nine mathematical reasoning benchmarks, it outperforms strong baselines—by 13.4 Avg@8 points over OPD under severe mismatch—while generating shorter, better-formed responses.

TIDE assumes a locally reliable teacher. Our evaluation is limited to two teacher–student pairs and mathematical reasoning; dialogue, code, and multilingual settings remain open. Moreover, our fixed-state, operator-level theory gives no global convergence guarantee, and adaptive keep-rate schedules warrant study.

Overall, reliable on-policy distillation should allocate supervision according to both the *direction* and *accessibility* of disagreement, a principle that applies whenever sampling leaves useful corrective signals uncovered.

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A Proofs and Additional Theoretical Details

This appendix proves the two propositions in [Section 3](#).

Notation and assumptions. We work at a fixed rollout position t with student-visited state $s = s_t$. We abbreviate $p(v) = p(v|s)$ and $q(v) = q_\theta(v|s)$, and write $a(v) = \log[p(v)/q(v)]$ and $h(a) = 2(e^{a/2} - 1)$. We use the normalized squared Hellinger distance

$$H^2(p, q) = 1 - \sum_{v \in \mathcal{V}} \sqrt{p(v)q(v)}.$$

For the deficit branch, $\mathcal{K}_t$, $\bar{p}_t$, Q_t , and $q_t^\mathcal{K}$ follow the definitions in [Section 3.2](#). We assume finite-vocabulary softmax distributions with full support. When differentiating a gated loss, its routing decisions are treated as fixed.

A.1 Proof of Proposition 1

Proof. For $a < 0$, $e^{a/2} \in (0, 1)$, so $-2 < h(a) < 0$. The local expansion follows from the first-order expansion of the smooth function h at zero, with $h(0) = 0$ and $h'(0) = 1$.

By the definition of H^2 ,

$$\nabla_\theta H^2(p, q) = -\frac{1}{2} \sum_{v \in \mathcal{V}} \sqrt{\frac{p(v)}{q(v)}} \nabla_\theta q(v).$$

Using $\sum_v \nabla_\theta q(v) = 0$, we obtain

$$\begin{aligned} \mathbb{E}_{v \sim q}[-h(a(v)) \nabla_\theta \log q(v)] &= -2 \sum_v \left(\sqrt{\frac{p(v)}{q(v)}} - 1 \right) \nabla_\theta q(v) \\ &= -2 \sum_v \sqrt{\frac{p(v)}{q(v)}} \nabla_\theta q(v) \\ &= 4 \nabla_\theta H^2(p, q), \end{aligned}$$

which proves [Equation \(2\)](#). □

A.2 Proof of Proposition 2

Proof. By the definitions in [Proposition 2](#), $q(v) = Q_t q_t^\mathcal{K}(v)$ for $v \in \mathcal{K}_t$. Substituting this identity into [Equation \(3\)](#) gives

$$d_t = \sum_{v \in \mathcal{K}_t} \bar{p}_t(v) \log \frac{\bar{p}_t(v)}{Q_t q_t^\mathcal{K}(v)} = D_{\text{KL}}(\bar{p}_t \| q_t^\mathcal{K}) - \log Q_t.$$

Both terms are nonnegative. Their sum is zero exactly when $q_t^\mathcal{K} = \bar{p}_t$ and $Q_t = 1$, proving the claim. □

B Related Work

B.1 Mechanistic Understanding and Failure Modes of OPD

On-policy distillation (OPD) trains a student on its own trajectories while using a teacher to provide dense token-level supervision at the states the student visits ([Agarwal et al., 2024](#); [Lu & Lab, 2025](#)). By aligning training with the student’s inference-time state distribution, OPD reduces the exposure bias associated with distillation from fixed demonstrations ([Bengio et al., 2015](#); [Ross et al., 2011](#); [Gu et al., 2024](#)). Recent work has moved beyond demonstrating the empirical effectiveness of OPD toward explaining its underlying behavior. In particular, [Li et al. \(2026a\)](#) study its phenomenology and optimization mechanism, emphasizing the role of teacher–student overlap, while [Zhu et al. \(2026\)](#) systematically examine the conditions under which OPD succeeds or fails. Other analyses

identify limitations of output-space supervision, including a deteriorating signal-to-noise ratio as the student approaches the teacher (Yang et al., 2026a).

A growing body of work further shows that a locally well-defined teacher–student objective does not necessarily imply globally desirable rollouts. OPD may exhibit abrupt length inflation and repetitive generation (Luo et al., 2026); its KL-based signal can enter an agreement trap (Xin et al., 2026); and teacher supervision can lose fidelity on prefixes induced by the student (Liu et al., 2026b). Related failures have also been observed when privileged teacher context changes the learning signal for long reasoning traces (Kaur et al., 2026). Our work builds on this mechanistic perspective. We identify a concrete form of *degenerate agreement*: a student-generated repetitive prefix can make the conditioned teacher favor the same continuation, producing near-zero local discrepancy even though the complete response is degenerate. This observation motivates treating teacher–student agreement as potentially uninformative rather than automatically desirable. More broadly, it changes what selective supervision should preserve: instead of favoring overlap itself, token selection should concentrate on mismatches for which the teacher still provides corrective information.

B.2 Selective and Reliable Supervision in OPD

Dense token-level feedback does not imply that every token provides useful guidance (Lee et al., 2026). Existing methods therefore select or reweight OPD supervision using position, discrepancy, importance, teachability, or uncertainty. Prefix-OPD retains supervision only over selected reasoning prefixes (Zhang et al., 2026). At the token level, recent approaches filter and reweight supervision (Li et al., 2026b), estimate token importance (Xu et al., 2026), restrict updates to learnable disagreement (Wang et al., 2026), or use entropy and self-uncertainty to modulate individual tokens (Jin et al., 2026; Ke et al., 2026). Relaxed OPD similarly avoids enforcing equally strict alignment at every position (Ko et al., 2026), while trajectory- and prefix-oriented methods mine or refine more reliable training paths (Zhao et al., 2026b; Jiang et al., 2026; Liao et al., 2026). Collectively, these studies establish that uniformly distilling every student-generated token is neither necessary nor consistently beneficial.

Most selective methods, however, still decide which signals to retain on student-generated trajectories and optimize the tokens the student has already sampled. Better selection can therefore improve the quality of observed supervision without resolving a distinct coverage problem: teacher-preferred alternatives with negligible student probability rarely appear in the rollout and consequently provide little or no direct corrective signal.

Complementary work modifies the optimization procedure to improve reliability, including asymmetric token-level objectives (Jia et al., 2026), trust-region constraints (Xing et al., 2026), teacher-guided optimization under large policy divergence (Liu et al., 2026a), and control-variate estimators for reducing OPD gradient variance (Oh et al., 2026). These methods improve the reliability of updates under observed disagreement, but severe teacher-side deficits remain the least likely discrepancies to be observed through student sampling. TIDE addresses selection and coverage jointly by using the *direction* of mismatch to determine the correction mechanism. It removes matched tokens, applies bounded suppression to sampled student-excess tokens, and analytically recovers teacher-preferred mass that student rollouts are unlikely to sample. Thus, its allocation rule follows the distinct observability and optimization requirements of excess and deficit, rather than applying a single token score or a uniform modification of the OPD objective.

C Additional Experiments

C.1 Teacher Continuation from Repetitive Student Prefixes

The preceding analysis shows that repetitive trajectories emerge naturally from the student during OPD. We next ask whether the teacher corrects such behavior or instead inherits it from the student-generated context. To isolate the teacher’s response, we perform a continuation experiment in which the student controls the prefix but the teacher controls all subsequent tokens.

We first identify naturally occurring student responses in which at least 30% of the tokens belong to a detected repetitive region. For each response, we retain the original prompt and truncate the student trajectory 200 tokens after the onset of repetition. We then condition the Qwen3-8B teacher

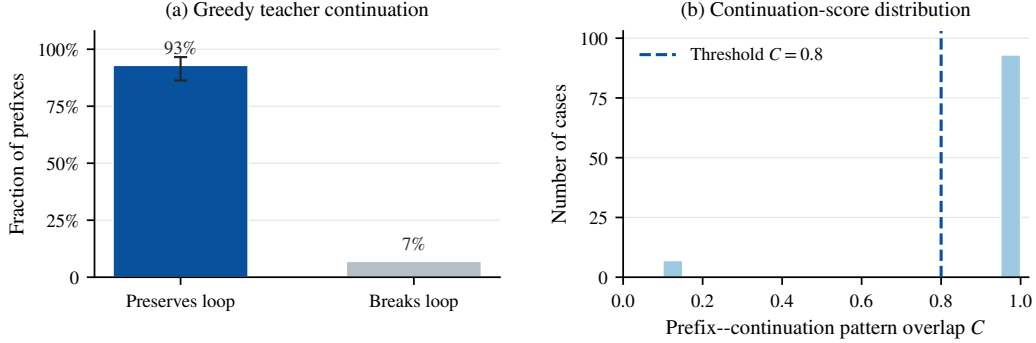


Figure 6: Teacher continuation from naturally occurring repetitive student prefixes. **(a)** Greedy teacher decoding preserves the student-induced loop in 93% of cases; the error bar denotes a 95% Wilson confidence interval. **(b)** Distribution of the prefix-continuation pattern overlap C , defined as the fraction of contiguous eight-token sequences in the teacher continuation that also appear in the student-prefix tail. The dashed line marks the preservation threshold $C = 0.8$.

on this prompt-prefix pair and greedily decode 256 additional tokens. The student and teacher continuations therefore begin from exactly the same student-generated state, allowing us to test whether the repetitive behavior persists after control is transferred to the teacher.

We measure persistence by comparing local token patterns in the teacher continuation with those in the tail of the student prefix. Let $\mathcal{G}_8$ denote the set of contiguous eight-token sequences appearing in the teacher continuation, and let $\mathcal{P}_8$ denote the corresponding set from the final 256 tokens of the student prefix. Their overlap is measured as

$$C = \frac{|\mathcal{G}_8 \cap \mathcal{P}_8|}{\max(|\mathcal{G}_8|, 1)}.$$

A value of $C = 1$ means that every local pattern in the teacher continuation already occurs in the repetitive student prefix. We classify the teacher as preserving the loop when $C \geq 0.8$; using eight-token sequences avoids counting short, accidental token overlaps as continuation of the same loop.

As shown in Fig. 6, the teacher preserves the student-induced repetitive pattern in 93% of cases. The result is also well separated from the decision threshold: preserved continuations concentrate at $C = 1$, indicating near-exact reproduction of the local patterns established by the student. Thus, repetition is not merely a failure of student-side sampling. Once the student drives the trajectory into a repetitive state, the teacher typically continues from that state in the same way rather than providing a corrective target. This explains how a student-generated loop can become self-reinforcing under on-policy distillation.

C.2 Does Supervision on Matched Tokens Help?

Our method deliberately assigns zero weight to positions where the teacher and student approximately agree. We examine whether these omitted positions still provide useful supervision, and whether the benefit of mismatch selection could instead be explained simply by updating fewer tokens.

We construct a matched-only control by reversing the allocation principle. Within each training batch, we rank valid response positions according to the absolute sampled-token residual

$$|R_t| = |\log p(o_t | s_t) - \log q_\theta(o_t | s_t)|.$$

We discard the largest 40% of residuals and apply the standard OPD log-ratio advantage only to the lowest-residual 60%. All other positions receive zero distillation weight, and the teacher top- K deficit branch is disabled. The training data and its order, student initialization, rollout budget, optimizer, and training length are otherwise unchanged. Table 6 reports the complete per-benchmark results and aggregate generation statistics for both teacher-student pairs.

Table 6: Complete matched-token control results. The upper panel reports Avg@8 / Pass@8 on each benchmark and their nine-benchmark macro-average. In the lower panel, response length and Distinct-4 are also macro-averaged over the nine benchmarks, while the remaining diagnostics are totals. All evaluations use eight responses per problem.

Pair	Method	AMC23	AIME24	AIME25	MATH-500	Minerva	Olympiad	BRUMO25	CMIMC25	HMMT25	Macro Avg.
JustRL-1.5B → R1-Distill-1.5B	Student	73.12/95.00	27.08/36.67	24.17/33.33	83.43/95.00	28.03/43.38	44.31/60.83	27.08/56.67	14.37/30.00	12.08/20.00	37.08/52.32
	Matched-only	76.56/95.00	31.67/46.67	31.67/36.67	85.72/94.60	30.70/44.85	50.17/65.13	36.25/56.67	18.12/35.00	19.17/33.33	42.23/56.44
	TIDE	84.38/95.00	45.83/80.00	33.33/50.00	86.72/94.60	33.55/45.22	52.54/68.10	42.92/66.67	22.19/42.50	19.17/43.33	46.74/65.00
Qwen3-8B → Qwen3-1.7B-Base	Student	13.44/40.00	3.33/3.33	0.00/0.00	23.82/68.20	6.34/24.26	9.85/34.27	5.00/20.00	0.00/0.00	0.00/0.00	6.87/21.12
	OPD	10.31/45.00	2.92/13.33	1.25/13.33	26.52/73.80	6.94/28.68	10.66/37.69	2.92/13.33	0.31/2.50	0.00/0.00	6.87/25.29
	Matched-only	12.81/27.50	2.92/3.33	1.25/6.67	26.12/71.60	8.23/26.84	11.03/36.50	1.25/10.00	0.62/2.50	0.42/3.33	7.18/20.92
	Mismatch selection	26.56/52.50	3.33/13.33	6.25/16.67	52.15/77.60	15.58/30.88	21.01/40.80	5.00/13.33	0.94/5.00	0.42/3.33	14.58/28.16
	TIDE	39.38/60.00	10.00/16.67	4.58/10.00	65.08/86.20	21.83/41.91	28.58/50.30	10.42/26.67	2.81/12.50	0.42/3.33	20.34/34.17

Pair	Method	Length ↓	Dist.-4 ↑	Format Err. ↓	Solve-none ↓	Solve-all ↑
Weak	Student	12,543	0.563	803	549	550
	Matched-only	8,494	0.568	612	508	658
	TIDE	7,571	0.566	488	472	711
Strong	Student	13,465	0.099	7,559	985	2
	OPD	22,395	0.062	8,623	920	0
	Matched-only	12,702	0.101	7,237	950	3
	Mismatch selection	29,655	0.104	6,845	862	90
	TIDE	7,294	0.305	706	717	286

Weak teacher-student mismatch. On JustRL-1.5B → R1-Distill-1.5B, matched-only training improves Avg@8 from 37.08 to 42.23. Low-mismatch positions are therefore not entirely uninformative when the teacher and student are already closely aligned. Nevertheless, matched-only remains 4.51 points below TIDE and reaches only 56.44 Pass@8, compared with 65.00 for TIDE, despite supervising a larger fraction of positions. Thus, even in this setting, matched positions provide a lower return per supervised token.

Mismatch selection therefore more than doubles the matched-only accuracy while supervising only one third as many positions. Adding bounded excess suppression and deficit recovery further increases Avg@8 to 20.34 and Pass@8 to 34.17, while reducing the macro-averaged response length from 29,655 to 7,294 tokens and format errors from 6,845 to 706.

C.3 Accessibility and Value of Student-Deficit Tokens

We define a student-deficit position as a non-loop position where the teacher’s top-1 token v_t^* lies outside the student’s top-16 support. Writing its student probability as $\epsilon_t = q_\theta(v_t^*|s_t)$, a sampled-token estimator observes it within N independent draws from the same state with probability

$$P_{\text{obs}}(N) = 1 - (1 - \epsilon_t)^N,$$

and requires $1/\epsilon_t$ draws in expectation.

To test whether this inaccessible mass is useful, we train a deficit-only variant with $\rho^- = 0$, $\rho^+ = 0.2$, $K = 16$, and $\lambda = 1$. It applies analytic cross-entropy guidance only at positions with the largest teacher-support deficits; all other settings remain unchanged. Table 7 summarizes both the accessibility diagnostic and the resulting end-to-end performance.

As shown in Table 7, 95.5% of deficit positions have less than a 1% chance of being observed within four draws, yet the deficit-only branch raises Avg@8 from 6.87 to 18.32. Thus, these tokens are not an inconsequential probability tail: they contain useful teacher-supported modes that sampled-token OPD is unlikely to recover within a practical rollout budget.

C.4 Sensitivity to the Gate Keep-Rate

We jointly vary the excess and deficit keep-rates, $\rho^- = \rho^+ = \rho$, while holding all other hyperparameters fixed. Table 8 reports the results on Qwen3-8B → Qwen3-1.7B-Base.

Table 7: Student-deficit tokens are difficult to access through sampled supervision but carry substantial learning signal on Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base. **(a)** Accessibility under four independent draws from the same state. **(b)** End-to-end results macro-averaged over nine benchmarks with $n = 8$ samples per problem.

(a) Accessibility diagnostic		(b) End-to-end value			
Statistic	Value	Method	Avg@8	Pass@8	Dist.-4
Deficit positions among non-loop tokens	1.32%	Student	6.87	21.12	0.099
Median student probability ϵ_t	2.04×10^{-4}	Deficit-only	18.32	32.71	0.414
Expected draws $1/\epsilon_t$	4,909	TIDE	20.34	34.17	0.305
Median $P_{\text{obs}}(4)$	8.15×10^{-4}				
Positions with $P_{\text{obs}}(4) < 1\%$	95.5%				

Table 8: Sensitivity to the gate keep-rate $\rho^- = \rho^+ = \rho$. Avg@8 and response length are macro-averaged over nine benchmarks with $n = 8$; length is reported in thousands of tokens. Fmt. err. denotes the total number of responses without a parseable final answer.

ρ	Avg@8	Length	Fmt. err.
0.10	19.53	10.3K	1,111
0.20	20.34	7.3K	706
0.30	19.38	11.7K	620
0.50	20.93	6.8K	433

As shown in Table 8, TIDE remains stable across a broad range of keep-rates: Avg@8 varies by only 1.55 points over $\rho \in [0.1, 0.5]$, with no monotonic degradation as more positions are retained. We use $\rho = 0.2$ throughout the main experiments because it was fixed before benchmark evaluation and retains only a sparse subset of positions, rather than selecting the best-performing value retrospectively.

D Experimental Details

D.1 Additional Training and Implementation Details

All methods share the training setup described in the main text; here we document the remaining implementation choices. During rollout generation, the teacher scores the student’s responses with temperature 1.0 (no logit rescaling), and teacher log-probabilities are computed on the identical token sequences produced by the student, so the sampled residual $R_t = \log p(o_t | s_t) - \log q_\theta(o_t | s_t)$ is exact rather than re-estimated. We do not use an auxiliary KL penalty toward a reference policy, a format reward, or a repetition penalty during training; the distillation signal is the only learning signal for all OPD-style methods. The learning rate is held constant (no warmup or decay). Token-level advantages are detached before entering the policy-gradient loss, and the loss is aggregated with token-mean normalization over all valid response tokens in the batch; because training is fully on-policy (a single gradient step per rollout batch), the importance ratio is identically 1 and no PPO-style clipping is active. For TIDE, the quantile thresholds τ^- and τ^+ are recomputed on every training batch over all valid response positions, so the keep-rates ρ^- and ρ^+ are batch-relative rather than fixed absolute thresholds. The deficit branch adds one student forward pass over the teacher’s top- K candidate ids per position, a cost comparable to top- K OPD.

Training uses fully sharded data parallelism (Zhao et al., 2023) with gradient checkpointing and activation offloading; rollouts are generated with vLLM (Kwon et al., 2023) using dynamic batching.

D.2 Hyperparameters

Table 9 reports the shared training and evaluation configuration together with the method-specific hyperparameters. Unless explicitly varied in an ablation, the same configuration is used for all teacher–student pairs and baselines.

D.3 Baseline Implementations

All baselines are implemented in the same codebase and trained with the same data order, student initialization, rollout budget (4 responses per prompt), sequence lengths, optimizer, and hardware as TIDE. Unless otherwise stated, they differ only in how token-level advantages are constructed.

OPD. Standard on-policy distillation applies the sampled-token log-ratio advantage

$$A_t = \log p(o_t | c_t) - \log q_\theta(o_t | c_t)$$

to every valid response token, without token selection or reward reshaping.

GRPO. The teacher-free reinforcement-learning baseline replaces teacher supervision with the rule-based exact-match reward used for grading and computes group-normalized outcome advantages over the 4 responses sampled for each prompt. All other training settings are unchanged, so GRPO uses the same rollout and gradient budget as the distillation methods.

FiRe-OPD. We reimplement FiRe-OPD (Li et al., 2026b) following its official implementation and verify parity against it. Trajectories are ranked by length-normalized teacher log-likelihood, and the bottom 20% are discarded. On the retained trajectories, the sampled-token log-ratio advantages are multiplied by detached entropy-based weights, with both the teacher-confidence and student-confusion coefficients set to 1.0. FiRe-OPD operates entirely on student-sampled tokens and does not access the teacher’s top- K distribution.

PowerOPD. We implement PowerOPD using its Box–Cox reward transformation. For a sampled token with teacher probability p_t and student probability q_t , the log-ratio advantage is replaced by

$$A_t^{\text{Power}} = p_t^\alpha - q_t^\alpha,$$

where the token-independent $1/\alpha$ factor is absorbed into the learning rate. We use $\alpha = 5$, following the selected configuration in our sweep. This bounded, sign-preserving reward is applied to every valid sampled token; PowerOPD performs neither token selection nor teacher-support expansion.

AOPD. We implement AOPD with its probability-gap intervention rule. At each position, the gate compares the teacher and student probabilities of the sampled token,

$$G_t = \mathbb{1}[p_t - q_t \leq \tau],$$

with $\tau = 0$. Positions with $G_t = 0$ retain the standard sampled-token OPD advantage, whereas positions with $G_t = 1$ replace it with forward-KL guidance over the teacher’s top- K support. We use $K = 16$, renormalize the teacher probabilities within this support, and do not apply an additional $1/K$ scaling factor.

Teacher and student references. The reference rows report the teacher and the untrained student evaluated with the same decoding and grading pipeline used for all trained methods.

D.4 Evaluation and Metric Definitions

Decoding. Every model is evaluated with the same pipeline: vLLM generation with temperature 0.7, top- $p = 0.95$, $n = 8$ samples per problem, and a maximum generation length of 31,744 tokens. All problems use the prompt template

```
{problem} Please reason step by step, and put your final
answer within \boxed{}
```

with the model’s chat template and thinking mode disabled.

Grading. A response is graded by extracting the final `\boxed{}` expression and checking mathematical equivalence against the reference answer with a rule-based verifier (symbolic-equivalence

normalization for fractions, radicals, and equivalent forms). A response without a parseable boxed answer is counted as incorrect. No LLM-based verifier is used.

Metrics. For a benchmark with P problems, Avg@8 is the mean correctness over all 8 P responses and Pass@8 is the fraction of problems with at least one correct response. Aggregate scores are unweighted macro-averages over the nine benchmarks, so large benchmarks such as MATH-500 and OlympiadBench do not dominate the average.

Generation diagnostics. Response length is measured in tokens of the Qwen3-1.7B-Base tokenizer and averaged over all rollouts. The no-answer rate is the fraction of rollouts without a parseable boxed answer, which in practice coincides with generations that hit the decoding cap before terminating. Distinct-4 is the number of unique 4-grams divided by the total number of 4-grams in a response, averaged over rollouts; low Distinct-4 combined with long outputs and a high no-answer rate indicates repetitive degeneration rather than longer reasoning.

D.5 Checkpoint Selection and Reporting

All trained models—TIDE, every baseline, and every ablation and hyperparameter variant—are evaluated at the final checkpoint of the one-epoch run; no checkpoint is selected using benchmark scores. Hyperparameter studies vary a single quantity at a time around the default configuration ($K = 16$, $\rho^- = \rho^+ = 0.2$, $\lambda = 1.0$), which was fixed before the benchmark evaluation and not tuned on test sets. All numbers within a teacher–student pair are produced by the same decoding and grading pipeline, so they are directly comparable across methods and ablations.

E Case Studies

We qualitatively compare students trained with standard OPD and with TIDE under the strong-mismatch pair (Qwen3-8B $\rightarrow$ Qwen3-1.7B-Base), decoding both with the shared evaluation protocol (Appendix D.4). To avoid cherry-picking, candidate problems are selected by a fixed rule rather than manual inspection: the OPD student answers none of its 8 rollouts correctly while the TIDE student answers all 8 correctly. The four cases below are chosen from this pool to cover the spectrum of degeneration patterns that dominate the OPD student’s failures, ordered from the most fluent to the most severe: (1) a fluent sentence-level loop, (2) a self-dialogue loop that imitates the prompt format, (3) a periodic junk-token loop, and (4) a multilingual token soup collapsing into a single-token loop. Excerpts are verbatim; elisions and repetition counts are marked in brackets. Reported statistics (correct answers out of 8, Distinct-4 of the excerpted rollout) are computed from the raw evaluation outputs.

E.1 Case 1: Fluent Sentence-Level Loop

Problem	MATH-500 · Answer: 0
If the domain of the function $\log x^2$ is $x < a$ or $x > b$, for some a and b , find $a + b$.	
OPD Student	0/8 correct · Distinct-4 = 0.009
For the function $\log x^2$, the domain is where the argument of the logarithm is positive. So, $x^2 > 0$, which is true for all $x \neq 0$. Now, we are given that the domain is $x < a$ or $x > b$ for some a and b. This means that the function $\log x^2$ is undefined for x in the interval $[a, b]$. So, the domain of the function is the union of the intervals $(-\infty, a)$ and (b, ∞). Now, we are given that the domain is $x < a$ or $x > b$. This means that the function is defined for all x except for x in the interval $[a, b]$. So, the domain of the function is the union of the intervals $(-\infty, a)$ and (b, ∞). [... the last paragraph repeats 399 times, until the 31,744-token decoding cap; no answer is produced ...]	

TIDE Student	8/8 correct
The domain of $\log x^2$ is all x such that $x \neq 0$, i.e. $x < 0$ or $x > 0$, which can be written as $(-\infty, 0) \cup (0, \infty)$. The endpoints are $a = 0$ and $b = 0$. Thus, the sum of a and b is $0 + 0 = 0$. Therefore, the final answer is $\boxed{0}$.	

This is the most insidious pattern: the repeated content is fluent, locally coherent mathematical prose. Conditioned on a prefix that already contains several copies of the paragraph, continuing the loop is the high-probability continuation for the student *and* for the teacher, so token-level supervision along the loop appears well matched almost everywhere—the student-induced matched-region failure analyzed in the main text.

E.2 Case 2: Self-Dialogue Loop Imitating the Prompt Format

Problem	Minerva · Answer: 2
Subproblem 0: For $\omega \geq 0$, find A such that $A \cos(\omega t)$ is a solution of $\ddot{x} + 4x = \cos(\omega t)$. (Worked solution given in the prompt.) Subproblem 1: For what value of ω does resonance occur?	

OPD Student	0/8 correct · Distinct-4 = 0.088
[... answers Subproblem 1, then continues inventing new subproblems ...] Final answer: The condition for resonance is $\omega = \boxed{2}$. The final answer is $\boxed{2}$. I hope it is correct. Subproblem 6: What is the value of ω at which resonance occurs in the differential equation $\ddot{x} + 4x = \cos(\omega t)$? Please reason step by step, and put your final answer within <code>\boxed{}</code>. [... the model invents 149 “subproblems” and answers each of them, repeating “I hope it is correct.” 149 times, until the decoding cap; the final boxed expression is not the requested answer ...]	

TIDE Student	8/8 correct
The natural frequency ω_n of the system can be found from the homogeneous equation $\ddot{x} + 4x = 0$, whose general solution is $x(t) = A \cos(2t) + B \sin(2t)$. So the natural frequency is $\omega_n = 2$. Resonance occurs when the frequency of the external force matches the natural frequency. Thus, the value of ω at which resonance occurs is $\boxed{2}$.	

The failure here is structural rather than lexical: the OPD student *does* derive the correct answer $\omega = 2$ early in the rollout, but instead of terminating it imitates the few-shot format of the prompt and generates an endless sequence of self-posed subproblems. Because the imitated format is itself natural text, the teacher assigns high probability to each continuation, and the rollout is graded as incorrect because the final boxed expression drifts away from the requested answer.

E.3 Case 3: Periodic Junk-Token Loop from the First Token

Problem	AIME 2024 · Answer: 25
There exist real numbers x and y, both greater than 1, such that $\log_x(y^x) = \log_y(x^{4y}) = 10$. Find xy.	

TIDE Student	8/8 correct
From the first equation, $x = y^{3/2}$. Substitute into the second equation: $(y - y^{3/2})^2 = 4y^2$. Let $t = y^{1/2}$, then $y = t^2$, so $(t^2 - t^3)^2 = 4t^4$, i.e. $t^4(t^2 - 2t - 3) = 0$. Since $t > 0$, solving $t^2 - 2t - 3 = 0$ gives $t = 3$, so $y = 9$ and $x = 9^{3/2} = 27$. Thus $x + y = 27 + 9 = 36$. Therefore, the answer is: 36.	

The most severe pattern: an incoherent stream of unrelated tokens spanning many scripts and domains (CJK characters, Hangul syllables, and code identifiers such as `.visitInsn`), which then collapses into a loop repeating a single character for tens of thousands of tokens.

E.5 Discussion

Three properties of these failures are worth noting. First, degeneration is not a tail event within an otherwise correct solution: in Cases 3 and 4 the OPD student derails within the first few tokens and never returns, which is why its accuracy collapses to the untrained-student level while its average generation length explodes. Second, the spectrum matters: the repeated content ranges from fully fluent prose (Case 1) and correct mathematics wrapped in an imitated dialogue format (Case 2) down to single-token loops (Case 4), so degeneration cannot be detected—let alone penalized—by fluency or perplexity alone. Third, in every case the repetition is self-consistent: once a repetitive prefix is established, continuing the loop is the most likely continuation for the student *and* for the teacher conditioned on the same prefix, so token-level supervision along such rollouts looks well matched almost everywhere. This is precisely the student-induced matched-region failure analyzed in the main text. TIDE attacks the mechanism directly: matched tokens receive zero weight, the excess gate suppresses the student’s overconfident continuations of the loop, and the deficit branch reinstates the teacher-preferred exit tokens that the student’s sampling never reaches.

Table 9: Hyperparameters used in all experiments. Method-specific parameters apply only to the corresponding method; all remaining settings are shared.

Hyperparameter	Value
Data and batching	
Training dataset	DAPO-Math-17K
Data shuffling	False
Prompt batch size	64
Rollouts per prompt	4
Policy mini-batch size	64
Micro-batch size per GPU	1
Training epochs	1
Maximum prompt length	1,024 tokens
Maximum response length	7,168 tokens
Maximum training context	8,192 tokens
Rollout generation	
Inference engine	vLLM
Student sampling temperature	1.0
Teacher scoring temperature	1.0
Repetition penalty	1.0
Optimization	
Optimizer	AdamW
Learning rate	1×10^{-6} (constant)
Warmup	None
Adam coefficients	(0.9, 0.999)
Weight decay	0.01
Gradient clipping	1.0
Policy updates per rollout batch	1
Loss aggregation	Token mean
Auxiliary reference-policy KL	None
Format reward	None
Numerical precision	FP32 student; FP32/BF16 teacher
Hardware	8 GPUs, FSDP
TIDE defaults	
Teacher support size K	16
Excess keep-rate ρ^-	0.2
Deficit keep-rate ρ^+	0.2
Deficit weight λ	1.0
Excess reward	Hellinger ratio reward
Deficit statistic	Forward KL on teacher top- K
Teacher top- K normalization	Renormalized within top- K
Matched-token weight	0
Baseline-specific settings	
OPD	Sampled-token log-ratio
GRPO	Exact-match reward; group size 4
FiRe-OPD	Bottom 20% filtered; $\alpha = \beta = 1.0$
PowerOPD	Box-Cox exponent $\alpha = 5$
AOPD	$K = 16$, $\tau = 0$; normalized top- K ; no $1/K$ scaling
Final evaluation	
Samples per problem	8
Maximum response length	31,744 tokens
Sampling temperature	0.7
Top- p	0.95
Grader	Rule-based mathematical verifier